

Opportunity Cost Aware Heuristics for the Weapon–Target Assignment Problem

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Abstract. The Weapon–Target Assignment Problem (WTAP) is a combinatorial optimization problem that seeks to allocate limited weapon resources to targets while minimizing their expected survival value. This work investigates an opportunity-cost-based heuristic that prioritizes targets according to the difference between their best and second-best currently feasible weapon assignments. This criterion captures the potential loss associated with delaying an assignment when a target has a particularly valuable weapon choice but a substantially weaker alternative. We develop deterministic and randomized opportunity-cost-based algorithms and explore perturbation-based strategies to diversify the search. Computational experiments are conducted across Uniform, Beta, and Gaussian kill-probability distributions and multiple problem sizes. Results show that the opportunity-cost criterion substantially improves upon conventional greedy assignment and produces solutions close to the optimal Hungarian assignment in many instances.

Keywords: Weapon–Target Assignment · Opportunity Cost · Randomized Algorithms · Combinatorial Optimization

1 Problem Statement

The Weapon–Target Assignment Problem (WTAP) is a fundamental resource-allocation problem in which a limited set of weapon resources must be assigned to a collection of targets. The objective is to determine an assignment that maximizes the effectiveness of the available resources while accounting for differences in weapon effectiveness and target importance. WTAP represents a challenging combinatorial optimization problem and has motivated the development of exact algorithms, greedy heuristics, metaheuristics, and other optimization techniques.

Consider a WTAP instance consisting of n weapon types and m targets. Let W_i denote the number of available units of weapon type i , and let $V_j > 0$ represent the importance or value of target j . The effectiveness of weapon type i against target j is represented by the kill probability

$$P_{ij} \in [0, 1].$$

The assignment decision is represented by a binary variable

$$D_{ij} = \begin{cases} 1, & \text{if weapon type } i \text{ is assigned to target } j, \\ 0, & \text{otherwise.} \end{cases}$$

For a given assignment, the probability that target j survives all weapons assigned to it is

$$\prod_{i=1}^n (1 - P_{ij})^{D_{ij}}.$$

Therefore, the expected total survival value of all targets is

$$C(D) = \sum_{j=1}^m V_j \prod_{i=1}^n (1 - P_{ij})^{D_{ij}}.$$

The objective of WTAP is to find an assignment matrix D that minimizes $C(D)$. A lower value of $C(D)$ therefore represents a more effective weapon–target assignment.

The assignment is subject to resource constraints. The number of assignments made using weapon type i cannot exceed its available capacity:

$$\sum_{j=1}^m D_{ij} \leq W_i, \quad i = 1, \dots, n.$$

In the formulation considered in this work, each target receives at most one weapon assignment:

$$\sum_{i=1}^n D_{ij} \leq 1, \quad j = 1, \dots, m.$$

Under this one-assignment-per-target setting, assigning weapon type i to target j reduces the expected survival value of that target by an amount proportional to

$$S_{ij} = P_{ij}V_j.$$

Thus, S_{ij} provides a natural measure of the immediate benefit of an individual weapon–target pairing. However, selecting assignments solely according to the largest value of S_{ij} does not account for the alternatives available to other targets. A locally attractive assignment may consume a weapon that represents a particularly important choice for another target whose remaining alternatives are substantially weaker. This competition for limited weapon resources motivates the opportunity-cost-based assignment strategy developed in this work.

2 Methods for Improving Greedy Assignment

We investigate several strategies for improving the quality of the baseline greedy solution. All methods use the same weapon–target effectiveness score,

$$S_{ij} = P_{ij}V_j, \tag{1}$$

where P_{ij} is the kill probability of weapon type i against target j , and V_j is the value of target j . The methods differ in how this score is used to determine the order in which

assignments are made. We begin with the baseline greedy method and then introduce an opportunity-cost criterion and randomized extensions designed to reduce the effect of locally greedy decisions.

2.1 Baseline Greedy Assignment

The baseline method follows the classical greedy weapon–target assignment algorithm (CWTA). It first constructs the score matrix

$$S = [S_{ij}], \quad S_{ij} = P_{ij}V_j. \quad (2)$$

At each iteration, the algorithm identifies the currently available weapon–target pair having the largest score,

$$(i^*, j^*) = \arg \max_{i,j} S_{ij}. \quad (3)$$

If weapon type i^* has remaining capacity, one unit is assigned to target j^* . The capacity of weapon type i^* is decreased by one, and target j^* is removed from further consideration. If a weapon type becomes unavailable, it can no longer participate in subsequent assignments. This procedure continues until no additional feasible assignments can be made.

The method is computationally simple and prioritizes the assignment with the largest immediate expected reduction in target survival value. However, it considers each weapon–target pair independently. Consequently, it does not consider how assigning a weapon to one target may affect the quality of the remaining choices available to other targets.

2.2 Opportunity-Cost-Based Assignment

To account for competition among targets for limited weapon resources, we introduce an opportunity-cost criterion. Rather than selecting the globally largest individual score, the method evaluates the quality of the alternatives currently available to each unassigned target.

For an unassigned target j , let

$$S_j^{(1)} = \max_{i: W_i^{\text{rem}} > 0} S_{ij} \quad (4)$$

denote its best currently feasible weapon score, where W_i^{rem} is the remaining capacity of weapon type i . Let

$$S_j^{(2)} = \max_{\substack{i: W_i^{\text{rem}} > 0 \\ i \neq i_j^{(1)}}} S_{ij} \quad (5)$$

denote its second-best feasible score. We define the opportunity cost of target j as

$$OC_j = S_j^{(1)} - S_j^{(2)}. \quad (6)$$

A large OC_j indicates that target j has a particularly valuable best assignment but a substantially weaker second-best alternative. If its preferred weapon becomes unavailable because it is assigned elsewhere, a large amount of assignment quality may therefore be lost.

The opportunity-cost heuristic gives priority to the target having the largest opportunity cost,

$$j^* = \arg \max_j OC_j, \quad (7)$$

and assigns that target its best currently feasible weapon,

$$i^* = \arg \max_{i: W_i^{\text{rem}} > 0} S_{ij^*}. \quad (8)$$

After the assignment, the corresponding weapon capacity is updated and the assigned target is removed. Because weapon availability changes after every decision, the best and second-best alternatives, and therefore the opportunity costs, are recomputed at each iteration.

This approach differs from the baseline greedy method in an important way. The baseline asks which feasible weapon–target pair provides the largest immediate score. The opportunity-cost method instead asks which target has the most to lose if its best currently available weapon is no longer available.

2.3 Randomized Opportunity-Cost Assignment

Although the opportunity-cost criterion incorporates information about alternative assignments, always selecting the target with the largest opportunity cost remains deterministic. An early decision can still influence all subsequent assignments. We therefore introduce a randomized opportunity-cost strategy that explores multiple assignment orders while preserving the opportunity-cost signal.

At each iteration, opportunity costs are computed for all currently unassigned targets and sorted in descending order. Instead of always selecting the single largest value, we construct a restricted candidate list containing the K targets with the highest opportunity costs,

$$C_K = \text{TopK}\{OC_j : j \text{ is unassigned}\}. \quad (9)$$

A target $j \in C_K$ is then selected according to

$$\Pr(j) = \frac{(OC_j + \epsilon)^\alpha}{\sum_{\ell \in C_K} (OC_\ell + \epsilon)^\alpha}, \quad (10)$$

where $\epsilon > 0$ prevents zero sampling weights and α controls the strength of the preference toward targets with larger opportunity costs.

Larger values of α make the procedure more strongly favor the highest-opportunity-cost target, whereas smaller values introduce greater exploration. Once a target has been

selected, it is assigned to its best currently feasible weapon. Opportunity costs are then recomputed before the next assignment.

The complete randomized construction is repeated using multiple independent random seeds. Every generated assignment is evaluated using the original WTAP objective $C(D)$, and the solution with the smallest survival value is retained.

2.4 Perturbed Randomized Opportunity-Cost Assignment

We further introduce small perturbations to diversify the assignment trajectories explored by the randomized method. For each independent construction, the weapon–target scores are slightly perturbed,

$$\tilde{S}_{ij} = S_{ij}(1 + \delta_{ij}), \quad (11)$$

where

$$\delta_{ij} \sim \mathcal{U}(-\rho, \rho) \quad (12)$$

and ρ controls the perturbation magnitude.

The perturbed scores are used only during solution construction. Small perturbations can alter the ordering of nearly equivalent weapon–target choices and therefore cause the algorithm to explore different assignment trajectories. This is particularly useful when several candidate assignments have similar scores.

Importantly, perturbation does not modify the WTAP instance or its objective. After an assignment has been constructed using $\tilde{S}$, its quality is always evaluated using the original kill probabilities P_{ij} , target values V_j , and objective $C(D)$. The best solution found across the unperturbed and perturbed runs is retained.

2.5 Hungarian Assignment Baseline

In addition to the greedy methods, we use the Hungarian algorithm as an optimal assignment baseline. Under the balanced setting considered in our experiments, the number of weapon types equals the number of targets ($n = m$), and each weapon type has one available unit. Thus, the assignment can be represented as a one-to-one matching between weapons and targets.

Using the same effectiveness score

$$S_{ij} = P_{ij}V_j, \quad (13)$$

the objective under the one-weapon-per-target constraint is equivalent to maximizing the total score of the selected assignments,

$$\max_D \sum_{i=1}^n \sum_{j=1}^m S_{ij} D_{ij}. \quad (14)$$

The Hungarian algorithm finds the optimal one-to-one assignment for this linear assignment problem in polynomial time. Since the standard Hungarian formulation is expressed as a minimization problem, we use

$$c_{ij} = -S_{ij} = -P_{ij}V_j \quad (15)$$

as the assignment cost.

Unlike the greedy and opportunity-cost-based methods, the Hungarian algorithm considers the assignment matrix globally rather than making a sequence of locally determined decisions. We therefore use its solution as a reference for evaluating the solution quality of the constructive heuristics. A heuristic solution closer to the Hungarian objective value indicates a smaller optimality gap under the balanced one-to-one problem setting.

3 Experimental Results

3.1 Dataset and Experimental Setup

We evaluate the proposed methods using 240 randomly generated WTAP instances covering three kill-probability distributions and four problem sizes. The number of weapon types is equal to the number of targets, with $n = m \in \{50, 75, 100, 125\}$. For each problem size, 20 independent instances are generated for each of three kill-probability distributions, resulting in $3 \times 4 \times 20 = 240$ test instances.

The three distributions are Uniform, Beta, and Gaussian. For the Uniform instances, kill probabilities are sampled from $\mathcal{U}[0, 1]$. For the Beta instances, probabilities are sampled from $\text{Beta}(2, 5)$, producing instances in which relatively low kill probabilities are common. For the Gaussian instances, probabilities are sampled from $\mathcal{N}(0.7, 0.15)$ and clipped to the interval $[0, 1]$. Target values are independently sampled from $\mathcal{U}[0.1, 10]$.

For every instance, we evaluate the original CWTAA greedy algorithm, the proposed opportunity-cost method (OC-CWTAA), randomized opportunity-cost variants, and the Hungarian assignment algorithm. The expected total survival value $C(D)$ is used as the primary performance measure; therefore, lower values indicate better assignments. Under the balanced one-to-one setting considered in these experiments, the Hungarian solution provides the optimal reference value.

3.2 Effect of Opportunity-Cost-Based Assignment

We first examine whether explicitly considering the opportunity cost of an assignment improves upon the baseline CWTAA greedy strategy. CWTAA selects the currently available weapon–target pair having the largest value of $P_{ij}V_j$, whereas OC-CWTAA prioritizes a target according to the difference between its best and second-best currently feasible weapon scores.

Table 1 compares the mean survival values obtained by CWTAA, OC-CWTAA, and the Hungarian baseline. Each entry represents the mean over 20 independently generated instances. Lower values indicate better solutions.

OC-CWTAA improves upon CWTAA on 239 of the 240 test instances. Averaged on a per-instance basis, introducing the opportunity-cost criterion reduces the survival objective by approximately 22.2% relative to CWTAA. The magnitude of the improvement varies across probability distributions. For the Beta instances, the improvement is

Table 1. Mean survival value of CWTAA, OC-CWTAA, and Hungarian over 20 instances. Lower is better.

Distribution	Size	CWTAA	OC-CWTAA	Hungarian
Uniform	50	9.560	7.848	7.260
Uniform	75	9.657	7.926	7.357
Uniform	100	9.770	7.726	7.150
Uniform	125	10.068	8.026	7.383
Beta	50	85.301	83.364	82.816
Beta	75	122.302	119.399	118.444
Beta	100	152.739	148.651	147.574
Beta	125	180.065	175.191	173.869
Gaussian	50	7.556	5.669	5.292
Gaussian	75	5.877	3.709	3.328
Gaussian	100	5.469	2.499	2.040
Gaussian	125	4.818	1.673	0.994

relatively small, while substantially larger reductions are observed for the Uniform and Gaussian instances.

The largest improvement occurs for the Gaussian instances and becomes more pronounced as problem size increases. At 125×125 , for example, the mean survival value decreases from 4.818 for CWTAA to 1.673 for OC-CWTAA, corresponding to approximately a 65.3% reduction. These results demonstrate that accounting for the quality of a target’s next-best alternative can substantially improve upon assignment decisions based solely on the immediate $P_{ij}V_j$ score.

3.3 Randomized Opportunity-Cost Assignment

Although OC-CWTAA substantially improves the baseline greedy solution, it remains deterministic and always selects the target with the largest opportunity cost. We therefore investigate whether controlled randomization among several high-priority targets can further improve solution quality.

The randomized opportunity-cost method (ROC-CWTAA) constructs a restricted candidate list containing the K_{RCL} targets with the highest opportunity costs. A target j in this list is selected with probability proportional to

$$(OC_j + \epsilon)^\alpha, \quad (16)$$

where α controls the selection pressure. Small values of α produce greater exploration, whereas larger values more strongly favor targets with high opportunity cost. The selected target is assigned its best currently feasible weapon, opportunity costs are recomputed, and the process continues until the assignment is complete. The entire construction is repeated using independent random seeds, and the solution with the lowest survival objective is retained.

We evaluate $K_{\text{RCL}} \in \{2, 5, 10\}$, $\alpha \in \{1, 2, 3, 5\}$, and search budgets of 10 and 20 randomized constructions. Table 2 compares deterministic OC-CWTAA with the best

Table 2. Mean survival values of deterministic and randomized opportunity-cost assignment. Lower is better.

Distribution	Size	OC-CWTAA	ROC-CWTAA	Hungarian
Uniform	50	7.848	7.475	7.260
Uniform	75	7.926	7.623	7.357
Uniform	100	7.726	7.460	7.150
Uniform	125	8.026	7.766	7.383
Beta	50	83.364	83.005	82.816
Beta	75	119.399	118.851	118.444
Beta	100	148.651	148.086	147.574
Beta	125	175.191	174.653	173.869
Gaussian	50	5.669	5.413	5.292
Gaussian	75	3.709	3.533	3.328
Gaussian	100	2.499	2.253	2.040
Gaussian	125	1.673	1.280	0.994

Table 3. Effect of selection-pressure parameter α . Values are mean percentage gaps from the Hungarian solution across the tested ROC-based configurations. Lower is better.

α	Budget 10	Budget 20
1	11.97%	9.97%
2	8.04%	6.81%
3	7.10%	6.07%
5	6.79%	5.94%

ROC-CWTAA result obtained for each distribution–size combination using a search budget of 20.

Randomization provides an additional improvement beyond the deterministic opportunity-cost criterion. When the best randomized configuration is selected for each instance, ROC-CWTAA improves upon OC-CWTAA on 236 of the 240 instances, with an average per-instance reduction in survival value of approximately 6.0%. The improvement is particularly noticeable for the larger Gaussian instances. At 125×125 , for example, the mean survival value decreases from 1.673 for OC-CWTAA to 1.280 for ROC-CWTAA.

To understand the behavior of the randomized method more closely, we next examine the effects of α and K_{RCL} . Rather than reporting only the best parameter combination, Table 3 reports the mean percentage gap from the Hungarian solution after pooling the tested ROC-based diversification configurations for each value of α .

A clear pattern is observed as α increases. With a search budget of 20, the average gap decreases from 9.97% at $\alpha = 1$ to 5.94% at $\alpha = 5$. A similar trend occurs with a budget of 10. Thus, completely broad exploration is less effective than maintaining a strong preference for targets with high opportunity cost. At the same time, the randomized procedure still allows alternative assignment sequences to be explored. Among the tested values, $\alpha = 5$ provides the best average performance.

Table 4 examines the size of the restricted candidate list. Here, K_{RCL} controls the number of high-opportunity-cost targets eligible for randomized selection at each step.

Table 4. Effect of restricted candidate-list size K_{RCL} . Values are mean percentage gaps from the Hungarian solution across the tested ROC-based configurations. Lower is better.

K_{RCL}	Budget 10	Budget 20
2	7.32%	6.55%
5	8.16%	6.80%
10	9.93%	8.24%

On average, smaller candidate lists perform better. At budget 20, $K_{\text{RCL}} = 2$ produces an average gap of 6.55%, compared with 6.80% for $K_{\text{RCL}} = 5$ and 8.24% for $K_{\text{RCL}} = 10$. This suggests that randomization is most effective when it remains concentrated among a small number of high-opportunity-cost targets. Increasing the candidate list introduces additional diversity, but it also increases the probability of selecting a target with a weaker opportunity-cost signal.

The effect of the search budget is also consistent across the tested parameters. Increasing the number of randomized constructions from 10 to 20 reduces the average gap for every value of α and K_{RCL} . For the unperturbed ROC-CWTAA method, the additional 10 constructions improve the best survival objective by approximately 0.99% on average. Thus, additional restarts provide a measurable improvement, although the gain is considerably smaller than the initial improvement obtained by introducing randomized opportunity-cost selection.

Overall, these experiments indicate that effective randomization requires a balance between exploitation and exploration. Larger α values preserve a strong preference for opportunity cost, while relatively small candidate lists prevent the search from becoming excessively random. The results therefore support using opportunity cost as the primary decision signal and randomization as a controlled mechanism for exploring alternative assignment sequences.

3.4 Perturbation-Based Randomization

We next investigate a second diversification strategy based on small random perturbations of the weapon–target scores. Randomized opportunity-cost selection changes the order in which high-priority targets are considered, but the underlying score matrix remains fixed. Perturbation provides an additional source of diversity by slightly modifying the scores used during solution construction.

For each randomized construction, the perturbed score is defined as

$$\tilde{S}_{ij} = S_{ij}(1 + \delta_{ij}), \quad (17)$$

where

$$\delta_{ij} \sim \mathcal{U}(-\rho, \rho) \quad (18)$$

and ρ determines the perturbation magnitude. We examine three small perturbation levels,

$$\rho \in \{0.01\%, 0.05\%, 0.10\%\}. \quad (19)$$

Table 5. Effect of score perturbation on randomized opportunity-cost assignment. Lower is better.

Distribution	Size	ROC-CWTAA	Perturbed ROC	Hungarian
Uniform	50	7.475	7.470	7.260
Uniform	75	7.623	7.642	7.357
Uniform	100	7.460	7.459	7.150
Uniform	125	7.766	7.751	7.383
Beta	50	83.005	82.955	82.816
Beta	75	118.851	118.803	118.444
Beta	100	148.086	148.081	147.574
Beta	125	174.653	174.580	173.869
Gaussian	50	5.413	5.368	5.292
Gaussian	75	3.533	3.517	3.328
Gaussian	100	2.253	2.225	2.040
Gaussian	125	1.280	1.260	0.994

Table 6. Effect of perturbation magnitude relative to matched unperturbed ROC-CWTAA configurations. Positive improvement indicates a reduction in survival value.

Perturbation	Mean Improvement	Configurations Improved
0.01%	0.049%	52.4%
0.05%	0.253%	58.0%
0.10%	0.351%	56.2%

The perturbed scores are used only to construct candidate assignments. Every resulting assignment is evaluated using the original kill-probability matrix P , target values V , and survival objective $C(D)$. Thus, perturbation changes the search trajectory but does not modify the problem being optimized.

Table 5 compares the best unperturbed ROC-CWTAA result with the best result obtained by incorporating score perturbation.

The improvement obtained from perturbation is smaller than that obtained from introducing opportunity cost or randomized target selection. Nevertheless, the results indicate that small perturbations can improve the search by changing the ordering of near-tied weapon–target scores.

To examine the magnitude of this effect, Table 6 compares the three perturbation levels directly. For each level, we match the perturbed ROC result with the corresponding unperturbed ROC configuration having the same problem instance, K_{RCL} , α , and search budget. The reported improvement is the percentage reduction in the best survival objective relative to that matched unperturbed ROC configuration.

The magnitude of perturbation has a measurable effect on solution quality. A perturbation of 0.01% produces only a 0.049% average improvement and improves approximately 52.4% of the matched configurations. Increasing the perturbation to 0.05% raises the average improvement to 0.253% and improves 58.0% of the configurations. The largest tested perturbation, 0.10%, produces the largest mean improvement of 0.351%, although its improvement frequency is slightly lower at 56.2%.

These results suggest that extremely small perturbations provide limited diversification because they rarely alter consequential assignment decisions. Increasing the

Table 7. Average percentage gap from the Hungarian solution for the ROC-CWTAA diversification strategies. Lower is better.

Strategy	Budget 10	Budget 20
ROC-CWTAA	8.71%	7.45%
ROC + Permutation	8.78%	7.43%
ROC + Perturbation	8.40%	7.13%
ROC + Perturbation + Permutation	8.36%	7.11%

magnitude to 0.05% or 0.10% is more effective at changing the ordering of closely scored alternatives while remaining sufficiently small to preserve the structure of the original score matrix. Among the tested values, 0.10% provides the largest average improvement, whereas 0.05% improves the largest fraction of configurations. Therefore, no single perturbation magnitude dominates according to both measures.

We also compare the overall randomized opportunity-cost families. With a search budget of 20, unperturbed ROC-CWTAA has an average gap of 7.45% from the Hungarian solution. Adding score perturbation reduces this average gap to 7.13%. Target permutation alone produces a similar gap of 7.43%, while combining perturbation and target permutation yields the smallest pooled gap of 7.11%.

The comparison shows that perturbation provides a more consistent benefit than target permutation alone. Target permutation changes the search trajectory but does not alter the underlying opportunity-cost information, and its average performance is nearly identical to unperturbed ROC-CWTAA. Perturbation, in contrast, directly changes near-tied score orderings and provides a modest but more consistent improvement. Combining perturbation and permutation produces the best pooled result, although its advantage over perturbation alone is small.

Taken together, these results indicate that the primary source of improvement remains the opportunity-cost criterion, followed by randomized selection among high-opportunity-cost targets. Score perturbation provides a smaller but measurable additional improvement, particularly at perturbation levels of 0.05% and 0.10%. Permutation contributes mainly as a secondary source of search diversity rather than as an independent improvement mechanism.

3.5 Overall Comparison

Finally, we compare the complete progression from the original CWTAA greedy method to the proposed opportunity-cost-based methods. To quantify solution quality relative to the optimal Hungarian reference, we define the percentage gap as

$$\text{Gap} = \frac{C_{\text{method}} - C_{\text{Hungarian}}}{C_{\text{Hungarian}}} \times 100\%. \quad (20)$$

A smaller gap indicates a solution closer to the Hungarian assignment, while a gap of zero indicates that the same objective value has been reached. In addition to CWTAA, OC-CWTAA, and ROC-CWTAA, we report the *Best OC Variant*. This value is not the result of a separate algorithm. For each problem instance, it represents the lowest survival

Table 8. Average percentage gap from the Hungarian baseline for CWTAA and the proposed opportunity-cost-based methods. Best OC Variant represents the best observed result across all tested randomized and diversified opportunity-cost configurations. Lower is better.

Distribution	CWTAA	OC-CWTAA	ROC-CWTAA	Best OC Variant
Beta	3.33%	0.73%	0.23%	0.15%
Uniform	34.13%	8.06%	3.01%	2.08%
Gaussian	195.10%	31.61%	9.72%	5.79%
Overall	77.52%	13.47%	4.32%	2.67%

objective obtained across all tested randomized opportunity-cost strategies, including randomized opportunity-cost assignment, score perturbation, target permutation, and the combination of perturbation and permutation. Thus, the Best OC Variant represents the best observed performance across the complete set of opportunity-cost-based diversification strategies evaluated in this study.

Table 8 reports the average per-instance percentage gap from the Hungarian solution for each probability distribution.

The results show a clear progression in solution quality. Across the complete benchmark, CWTAA has an average gap of 77.52% from the Hungarian reference. Introducing the opportunity-cost criterion reduces this gap substantially to 13.47%. Randomized opportunity-cost search further reduces the average gap to 4.32%. When the best solution obtained across all tested opportunity-cost-based diversification strategies is considered, the average gap decreases further to 2.67%.

The magnitude of the improvement varies across the three probability distributions. For the Beta instances, the Best OC Variant achieves an average gap of only 0.15% from the Hungarian solution. For the Uniform instances, the corresponding gap is 2.08%. Gaussian instances remain the most challenging, with an average gap of 5.79%. Nevertheless, this represents a substantial improvement over CWTAA, whose average gap is 195.10% for the same distribution.

Overall, the results indicate that the largest improvement is obtained by changing the assignment criterion itself. Replacing the immediate greedy $P_{ij}V_j$ decision with the proposed opportunity-cost criterion reduces the overall gap from 77.52% to 13.47%. Controlled randomization provides a further substantial improvement by exploring alternative assignment sequences. Score perturbation, target permutation, and their combination provide additional diversification and further reduce the best observed gap. These results demonstrate that opportunity-cost information provides the primary improvement over the baseline greedy strategy, while randomized diversification progressively moves the resulting assignments closer to the Hungarian reference.